

GCHandle.Free Method (System.Runtime.InteropServices)

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Content

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[]()

GCHandle.Free Method

Definition

Namespace: [System.Runtime.InteropServices](#) Assemblies:mscorlib.dll, System.Runtime.InteropServices.dll Assemblies:netstandard.dll, System.Runtime.dll
Assembly:System.Runtime.InteropServices.dll Assembly:System.Runtime.dll
Assembly:mscorlib.dll Assembly:netstandard.dll Source:[GCHandle.cs](#) Source:[GCHandle.cs](#)
Source:[GCHandle.cs](#) Source:[GCHandle.cs](#) Source:[GCHandle.cs](#)

Important

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Releases a [GCHandle](#).

```
public:
    void Free();
```

```
[System.Security.SecurityCritical]
public void Free();
```

```
public void Free();
```

```
[<System.Security.SecurityCritical>]
member this.Free : unit -> unit
```

```
member this.Free : unit -> unit
```

```
Public Sub Free ()
```

Attributes

[SecurityCriticalAttribute](#)

Exceptions

[InvalidOperationException](#)

The handle was freed or never initialized.

Examples

The following example shows an `App` class that creates a handle to a managed object using the `GCHandle.Alloc` method, which prevents the managed object from being collected. A call to the `EnumWindows` method passes a delegate and a managed object (both declared as managed types, but not shown), and casts the handle to an `IntPtr`. The unmanaged function passes the type back to the caller as a parameter of the callback function.

```

using System;
using System.IO;
using System.Threading;
using System.Windows.Forms;
using System.Runtime.InteropServices;

public delegate bool CallBack(int handle, IntPtr param);

internal static class NativeMethods
{
    // passing managed object as LPARAM
    // BOOL EnumWindows(WNDENUMPROC lpEnumFunc, LPARAM lParam);

    [DllImport("user32.dll")]
    internal static extern bool EnumWindows(CallBack cb, IntPtr param);
}

public class App
{
    public static void Main()
    {
        Run();
    }

    public static void Run()
    {
        TextWriter tw = Console.Out;
        GCHandle gch = GCHandle.Alloc(tw);

        CallBack cewp = new CallBack(CaptureEnumWindowsProc);

        // platform invoke will prevent delegate to be garbage collected
        // before call ends

        NativeMethods.EnumWindows(cewp, GCHandle.ToIntPtr(gch));
        gch.Free();
    }

    private static bool CaptureEnumWindowsProc(int handle, IntPtr param)
    {
        GCHandle gch = GCHandle.FromIntPtr(param);
        TextWriter tw = (TextWriter)gch.Target;
        tw.WriteLine(handle);
        return true;
    }
}

```

}

}

```

Imports System.IO
Imports System.Threading
Imports System.Windows.Forms
Imports System.Runtime.InteropServices
Imports System.Security.Permissions

Public Delegate Function Callback(ByVal handle As Integer, ByVal param As IntPtr) As Boolean

Friend Module NativeMethods

    ' passing managed object as LPARAM
    ' BOOL EnumWindows(WNDENUMPROC lpEnumFunc, LPARAM lParam);
    <DllImport("user32.dll")>
    Friend Function EnumWindows(ByVal cb As Callback, ByVal param As IntPtr) As Boolean
    End Function
End Module

Module App

    Sub Main()

        Run()

    End Sub

    <SecurityPermission(SecurityAction.Demand, UnmanagedCode:=True)>
    Sub Run()

        Dim tw As TextWriter = Console.Out
        Dim gch As GCHandle = GCHandle.Alloc(tw)

        Dim cewp As Callback
        cewp = AddressOf CaptureEnumWindowsProc

        ' platform invoke will prevent delegate to be garbage collected
        ' before call ends
        NativeMethods.EnumWindows(cewp, GCHandle.ToIntPtr(gch))
        gch.Free()

    End Sub

    Function CaptureEnumWindowsProc(ByVal handle As Integer, ByVal param As IntPtr) As Boolean

```

```
Dim gch As GCHandle = GCHandle.FromIntPtr(param)
Dim tw As TextWriter = CType(gch.Target, TextWriter)
tw.WriteLine(handle)
Return True
```

```
End Function
```

```
End Module
```

Remarks

The caller must ensure that for a given handle, [Free](#) is called only once.

Applies to

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